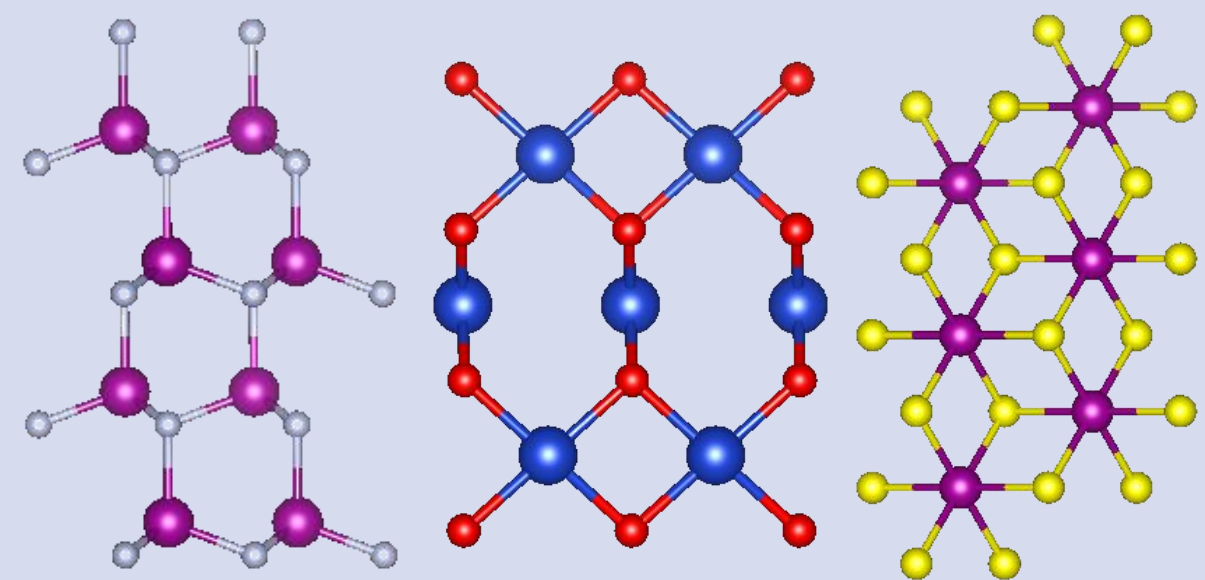
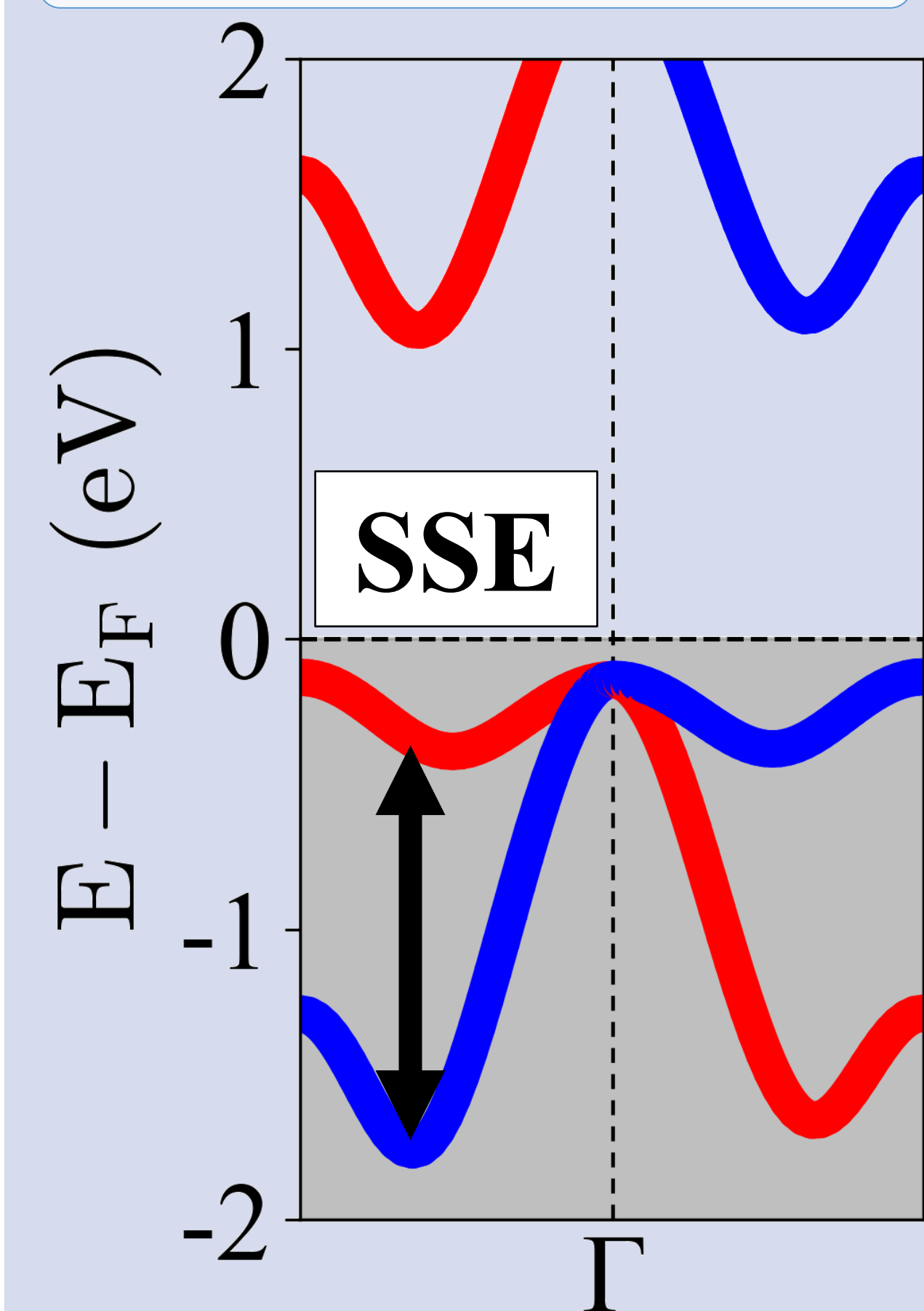


Generation

MatterGen
structure
generation



DFT-computed
SSE labels



SSE: Max spin
splitting in
 $[E_F - 2 \text{ eV}, E_F]$

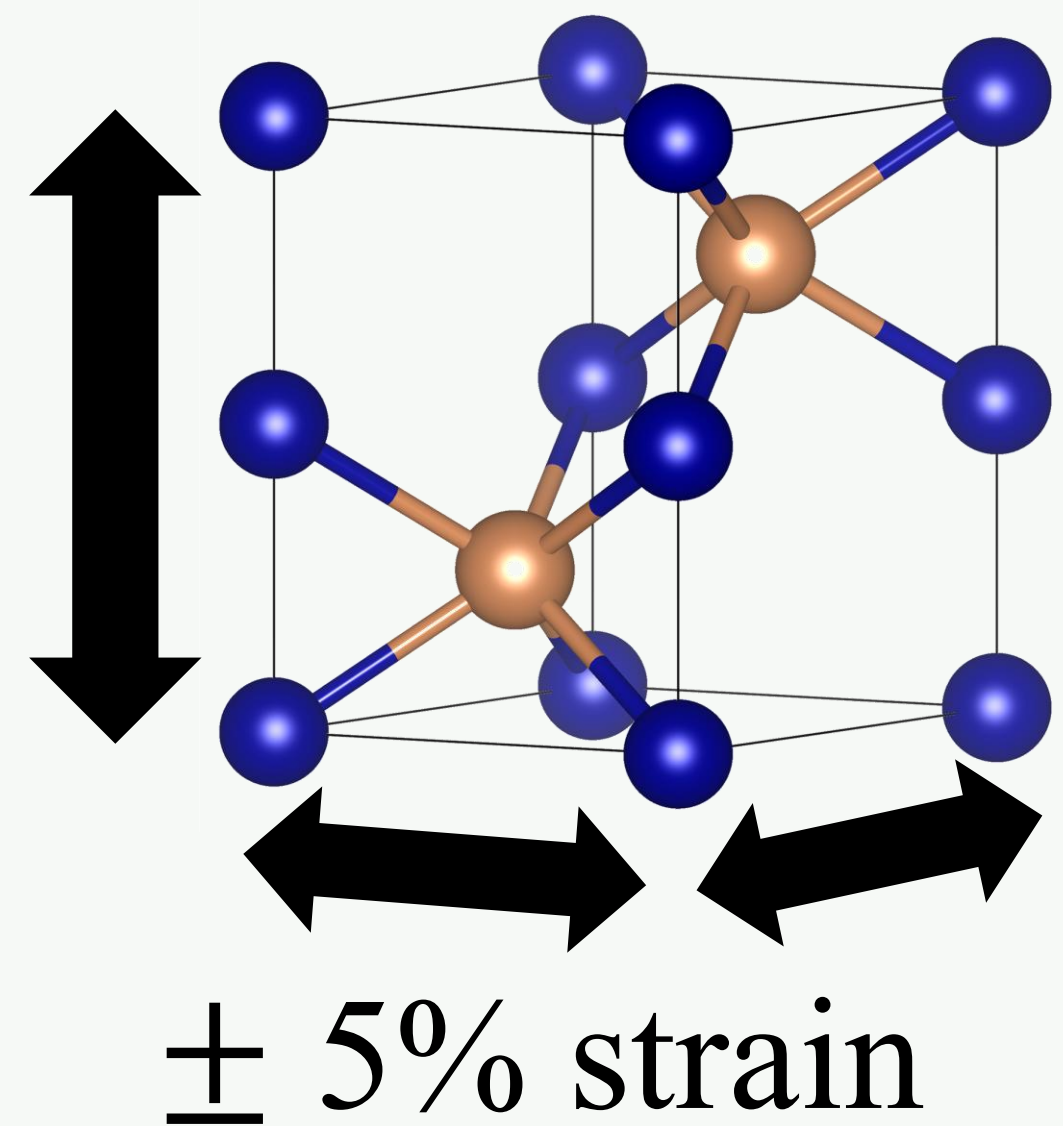
Screening

Binary
altermagnet
screening

Criterion
 $|M| < 0.005 \mu_B$
&
 $\text{SSE} \geq 1 \text{ meV}$

322 parents

Strain sampling



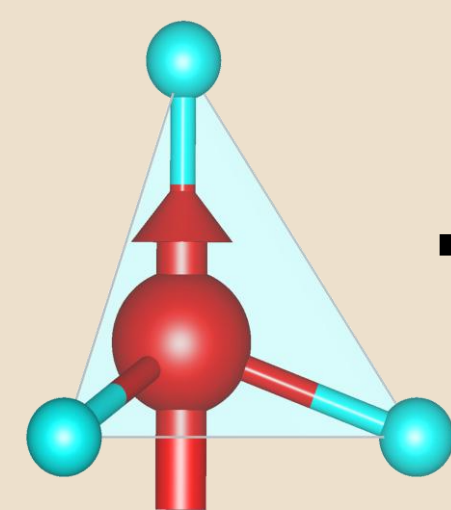
$\pm 5\%$ strain

N = 3,845

Modeling

Motif-based
descriptors

Motif



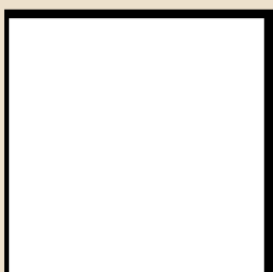
MSBI
MPF
 $XM X_{\text{std}}$
 Z_M
...

52 DFT-free
descriptors

XGBoost
surrogate for SSE
SHAP
interpretation

MSBI

MPF

 p/d ratio

|SHAP value|

Optimization

Descriptor-space
Bayesian
optimization

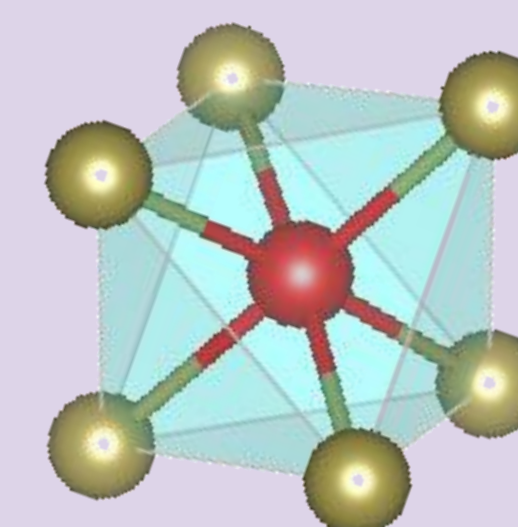
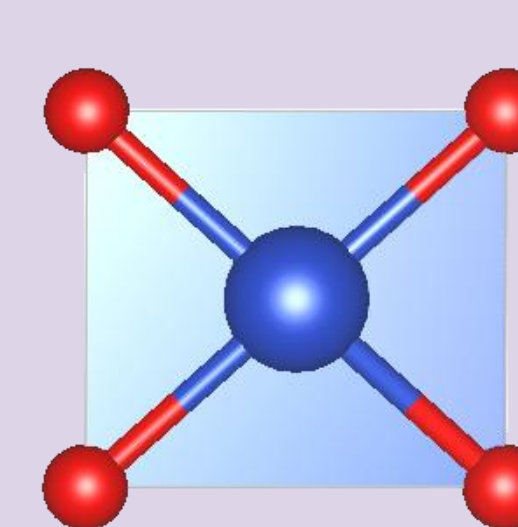
Optimization loop

Sampled
descriptors
↓
XGBoost
surrogate
↓
Maximize
predicted SSE
↓
Top candidates

Prototype matching

CN = 4

CN = 6

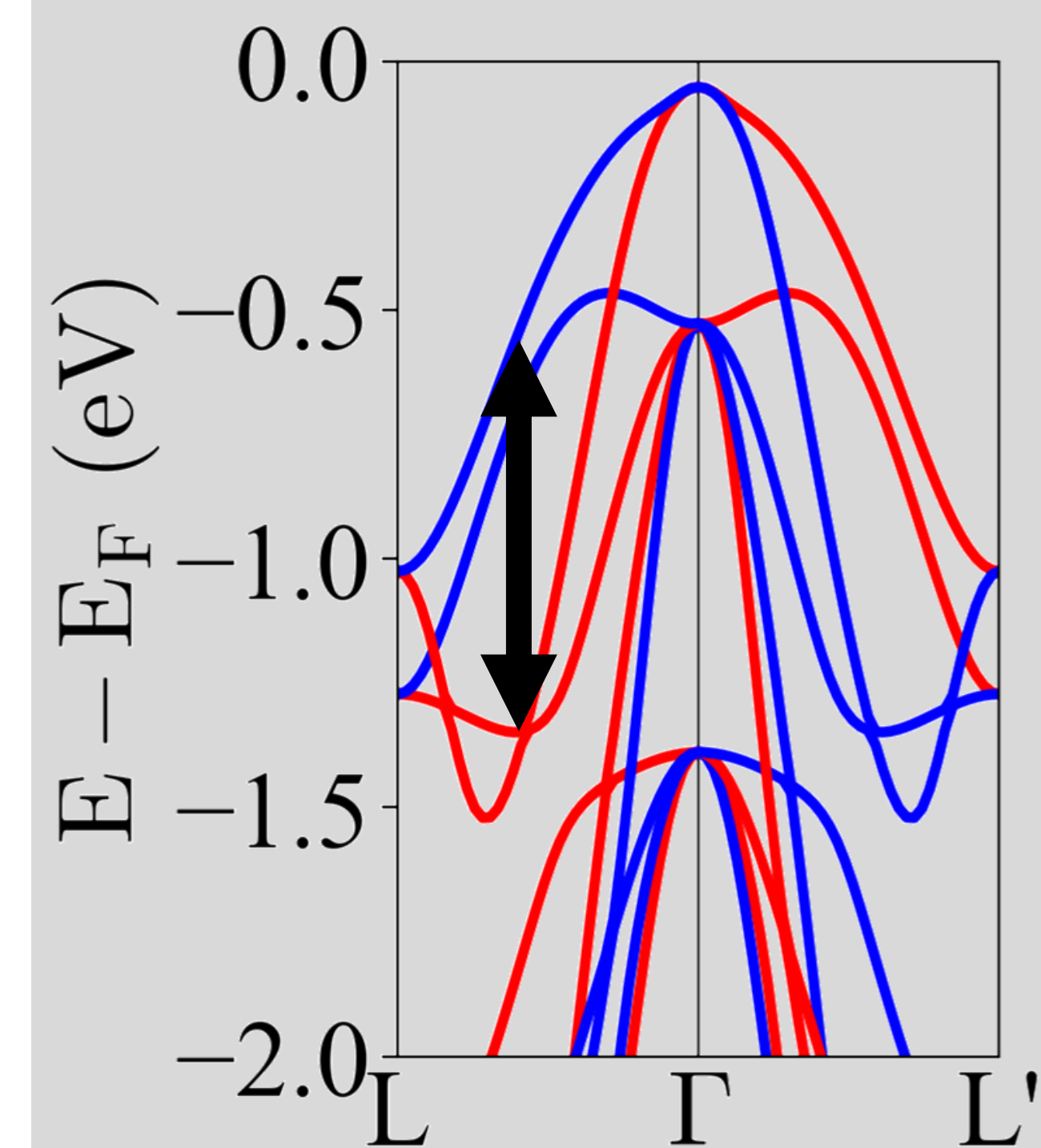
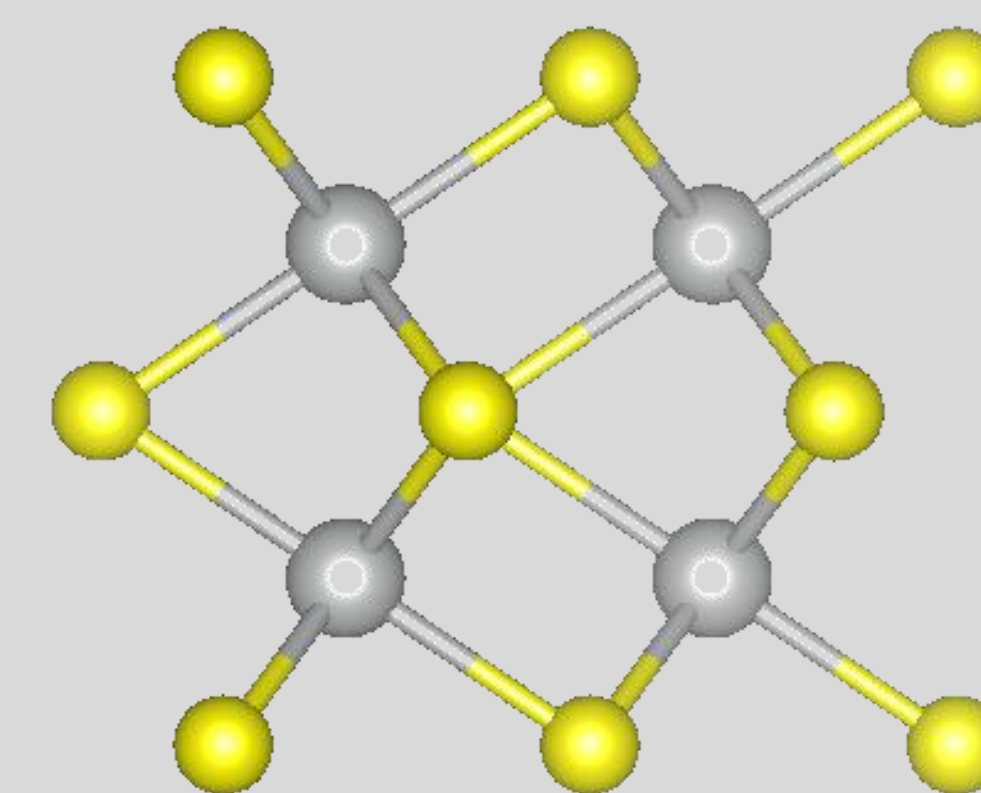


Validation

DFT validation

Top-5 predicted
candidates
($\text{CN} \in \{4, 6\}$)

High SSE material
 α -NiS identified



SSE: 0.823 eV